Name: Md. Jubayer Alam
ID: IT24649

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Q

Is set of odd numbers with binary operations (+), i.e. $\langle O, + \rangle$ an abelian group? If not explain the reasons with necessary notations

Ans

Let, $O = \{\ldots, -3, -1, 1, 3, 5, \ldots\}$

the set of all odd integers and consider the binary operation +.

Now, check whether $(O, +)$ forms an abelian group.

(i) Closure: Any two number $a, b \in O$

let, $a = 3$, $b = 5$

Now $a+b = 3+5 = 8$,

but 8 is not from O [Even]

$\therefore$ closure property doesn't match.

(ii) Associativity: If $\forall\, a, b, c \in O$ then

$a+(b+c) = (a+b)c \in O$

let, $a = 3$, $b = 5$, $c = -5$

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then, $3+(5-5) = (3+5)-5 = 3 \in O$

$\therefore$ Associativity matches.

(iii) Identity Element

A group requires identity ~~ser~~ $e \in O$

such that: $a+e=a \quad \forall a \in O$

The additive identity for integers is 0.

but $0 \notin O$ [Even]

$\therefore$ No identity element in O.

(iv) Inverse Element

Each $a \in O$ must have inverse $b \in O$ such that:

$$a+b=0$$

let, $a=3$, $b=-3$

$\therefore a+b = 3-3 = 0$

so inverse exists in O.

(v) Commutarity

Addition is commutative

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$a+b = b+a \quad \forall a, b \in O$

let, $a = 5$ $\quad a+b = 5+7 = 12$

$b = 7$ $\quad b+a = 7+5 = 12$

but $12 \notin O$

commutarity satisfy but 12 is not from O.

so $(O, +)$ is not an abelian group.

1. closure fails
2. No identity element
3. Inverse property fails without identity.

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